Intelligent Ranking & Chat System for Property Rental

Contents

Introduction & Background 01

Objectives & Scope 02

System Analysis & Literature Survey 03

System Architecture & Design 04

Implementation & Testing 05

Feasibility Study & Future Scope 06

Closing 07

Rental Market Challenges

Urban migration and rental competition

Rapid urban migration intensifies rental demand, increasing competition and driving up prices, complicating tenant searches and necessitating intelligent solutions to optimize property selection and ranking.

Issues with outdated and unverified listings

Outdated and unverified listings lead to misinformation, reduced trust, and inefficient property searches, complicating decision-making and decreasing overall rental market transparency and user satisfaction.

Manual search inefficiencies

Manual property searches are time-consuming, often yielding irrelevant results due to limited filtering options, leading to inefficient decision-making and increased user frustration in the rental market.

Need for intelligent automation

The rental market faces inefficiencies and information overload, necessitating intelligent automation to enhance property matching, streamline decision-making, and improve user experience through data-driven ranking and interaction systems.

Market Needs & Problem Statement

Limitations of current platforms

Current platforms lack personalized ranking, offer limited contextual chat support, and fail to integrate dynamic user preferences, resulting in inefficient property search and suboptimal rental matching experiences.

Intelligent Ranking System

The intelligent ranking engine analyzes sentiment, recency, reliability, and user preferences to deliver personalized, data-driven recommendations, ensuring users quickly find relevant and high-quality rental listings.

Real-Time Chat & Communication

The real-time chat module, built using Socket.io and Redis, enables instant, secure communication between tenants and owners, reducing delays, improving clarity, and enhancing decision-making during the rental process.

Project Objectives

01

Building a trustworthy, personalized rental platform

Develop a reliable rental platform leveraging AI for personalized property ranking and interactive chat, enhancing user trust through accurate recommendations and secure data handling.

02

Machine learning-based dynamic ranking

Build an intelligent ranking system using sentiment analysis, recency decay, confidence scoring, and user feedback to dynamically optimize listings and deliver personalized, reliable rental recommendations.

03

Secure in-platform communication

Ensure encrypted, authenticated communication within the platform to protect user data, prevent unauthorized access, and maintain confidentiality during property rental interactions.

Functional Goals & User Scope

01 FIRST

Personalized recommendations & sentiment analysis

Deliver personalized property recommendations using user preferences and behavior data while integrating sentiment analysis to enhance ranking accuracy and user satisfaction within the rental platform, ensuring smarter decision-making and a more intuitive, seamless user experience throughout the search process.

02 SECOND

Real-time chat with history and security

Enable secure, real-time chat with complete message history, ensuring encrypted data transmission and user authentication to maintain privacy and integrity within the property rental platform, while providing a smooth communication flow that helps users clarify details, negotiate terms, and make confident rental decisions.

03 THIRD

User groups: Renters, landlords, admins

Renter, landlord, and admin user groups are supported; renters seek tailored property matches, landlords manage listings efficiently, and admins oversee system integrity and user interactions, ensuring the platform operates smoothly, maintains reliable data flow, and delivers a seamless experience for all users involved.

AI Techniques Overview

Sentiment analysis

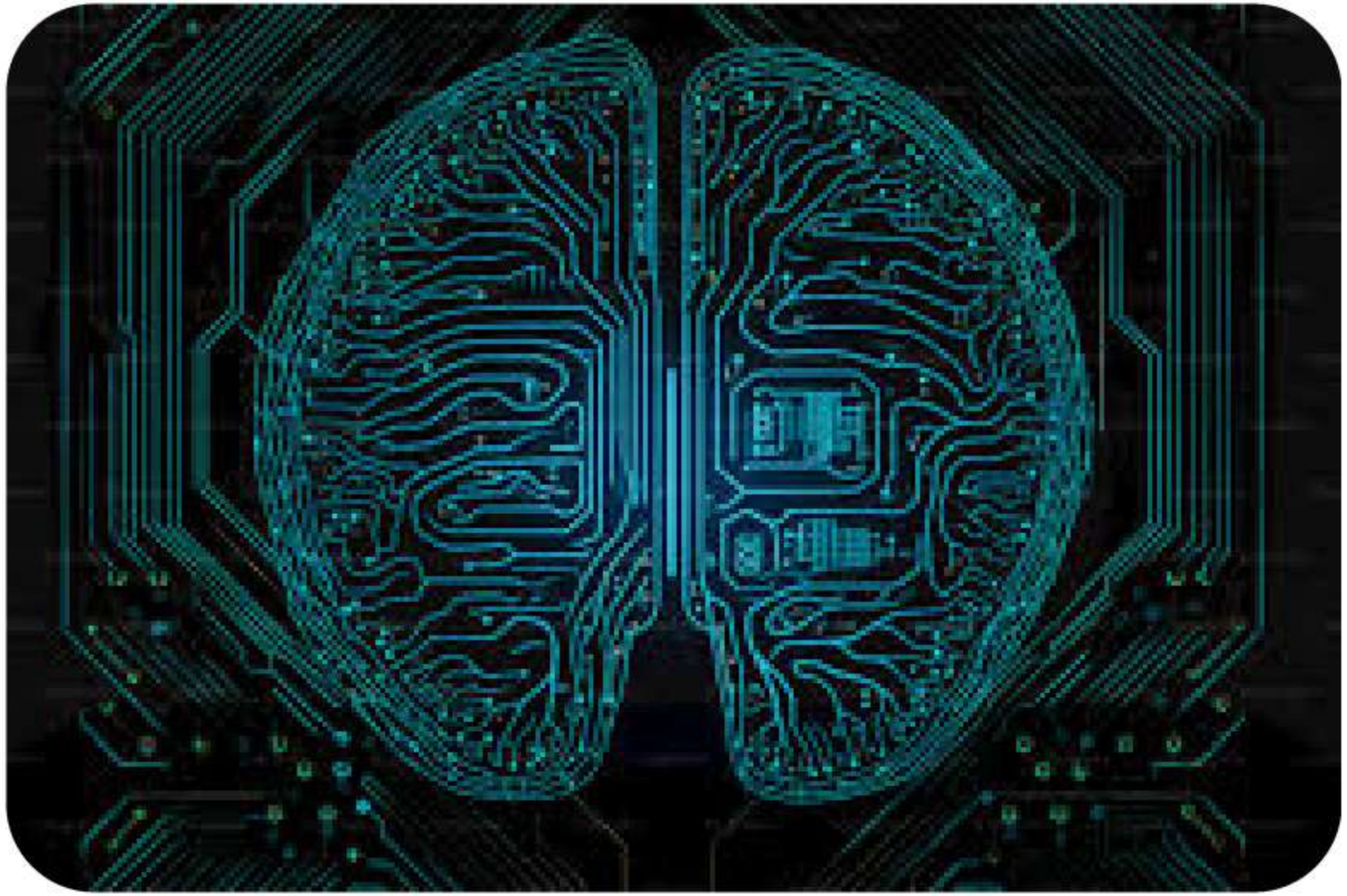
Analyzes user feedback, chat interactions, and reviews to extract sentiment cues, refining property ranking scores and improving the system's understanding of user intent.

VADER model for real-time feedback

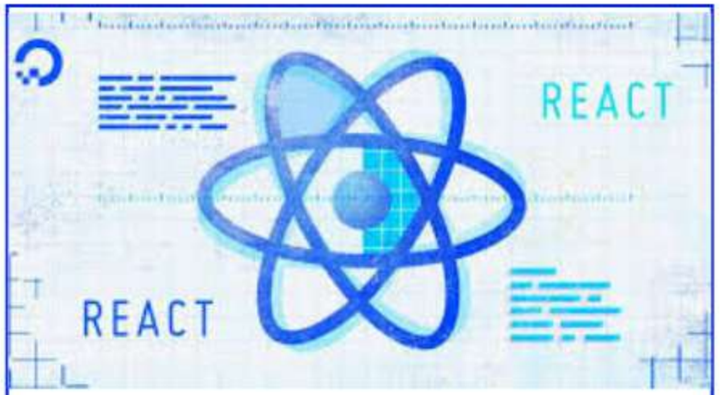
VADER is a lexicon and rule-based sentiment analysis tool optimized for real-time social media text, providing accurate polarity scores essential for evaluating user feedback in property rental applications.

Dynamic Ranking Model

Uses machine learning algorithms to continuously evaluate property features, user behavior patterns, and market signals to rank listings with high contextual accuracy.



Overall System Architecture



Frontend: React.js responsive UI

The frontend employs React.js to create a responsive user interface, ensuring seamless interaction across devices. It efficiently manages state and renders components dynamically for real-time updates.

Backend: Spring Boot for APIs

The backend utilizes Spring Boot to develop RESTful APIs, ensuring robust security through integrated authentication and authorization mechanisms, enabling efficient data handling and secure communication within the system.

AI Module: Python Flask services

The AI module leverages Python Flask to build lightweight RESTful services, while Redis is used for high-speed caching and session management, enabling efficient execution of ranking algorithms within a modular, scalable, and maintainable architecture.

Backend and Storage Design

Authentication and role-based access

Authentication uses secure token-based methods ensuring user identity verification, while role-based access control restricts permissions according to user roles, maintaining data integrity and system security in backend operations.

MySQL and Redis caching strategy

MySQL handles persistent storage and complex queries for property data, while Redis provides in-memory caching to accelerate frequent searches and reduce database load, ensuring low-latency responses.

Chat message storage and retrieval

Chat messages are stored in a NoSQL database optimized for fast writes and scalability. Retrieval employs indexed queries supporting real-time access and efficient thread reconstruction for seamless conversation flow.

```
@Controller
public class ChatController {

    private final SimpMessagingTemplate messaging
    private final ChatService chatService;

    public ChatController(SimpMessagingTemplate m
        this.messagingTemplate = messagingTemplat
        this.chatService = chatService;
    }

    @MessageMapping("/private-message")
    public void sendPrivateMessage(@Payload ChatM
        ChatMessage savedMessage = chatService.sa

        messagingTemplate.convertAndSend(
                "/user/" + chatMessageDto.getRece
                chatMessageDto
        );
    }
}
```

AI Module Components

01

Ranking logic using price, locality, amenities

The AI module ranks properties by integrating price, locality, and amenities through a weighted scoring algorithm, optimizing relevance for accurate rental recommendations, while continuously refining results behavioral insights to deliver smarter suggestions over time.

02

Sentiment processing with Wilson + Freashness Score

Sentiment processing utilizes natural language understanding combined with Wilson Score and Freshness Score to evaluate user feedback more reliably, applying continuous learning to dynamically refine sentiment models for higher accuracy and responsiveness in property ranking and chat interactions.

Frontend & Backend Implementation

Search filters and listing cards

Search filters enable dynamic property refinement based on user preferences, while listing cards present summarized property details with images and key attributes, ensuring efficient frontend-backend data synchronization and responsive interaction.

Secure login and listing management

Implemented secure login using JWT for session management and encrypted password storage. Listing management includes role-based access control, ensuring only authorized users can create, update, or delete property listings.

Document Verification

The document verification module uses OCR and machine-learning–based classifiers to extract and validate tenant documents with high accuracy. It detects tampering through metadata analysis, pattern inconsistencies, and fraud-risk scoring. This ensures fast, reliable, and real-time approval while reducing manual review efforts.

Real-time chat engine with WebSocket

The real-time chat engine utilizes the WebSocket protocol to enable persistent, bidirectional communication between client and server, ensuring low-latency message exchange and seamless user interaction within the frontend and backend integration, while maintaining stable connections, reducing overhead, and supporting smooth conversational flow even during peak usage.

Testing Strategies

Unit testing for backend, frontend, AI models

Unit testing ensures backend APIs function correctly, frontend components render as expected, and AI models produce accurate rankings, enabling early defect detection and robust system performance validation.

Integration testing for workflow validation

Integration testing verifies seamless interaction between ranking and chat modules, ensuring accurate data flow and consistent property rental workflows across the system.

Performance and load testing with caching

Performance testing evaluates system responsiveness under typical load, while load testing stresses the system with high user demand; caching optimizes response time by reducing database queries and server load.

Conclusion & Future Enhancements

Advanced Chatbot Integration

Future updates will include a more intelligent chatbot capable of multi-turn reasoning, proactive suggestions, and deeper context retention, enabling tenants and owners to complete queries, negotiations, and support tasks seamlessly within the platform.

Deep Personalization Engine

The platform will evolve to deliver hyper-personalized recommendations using behavioral clustering, adaptive preference learning, and continuous feedback loops, ensuring users receive property matches that align more precisely with lifestyle, budget, and long-term needs.

Automated verification and payment integration

Automated tenant verification enhances reliability by reducing fraud, while payment integration streamlines transactions, improving user experience and enabling seamless rental processing within the intelligent system.

Advanced machine learning personalization

Incorporating advanced machine learning personalization can enhance user experience by dynamically adapting property recommendations based on individual preferences and behaviors, improving relevance and increasing system efficiency.

Thanks